Identifying and Rejecting Backgrounds from Through-Going Beam Muons for Muon Capture at Rest Signal in ProtoDUNE-HD

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Outline

Muon capture at rest (μCAR) studies have been actively conducted, including ongoing efforts by the South Dakota Mines group focusing on event selection.

The work primarily focus on understanding the background contribution of blips from through going beam muons to those from muon capture at rest process (muCAR).

- Understanding muCAR
- Reconstructed beam event selection
- Through going beam muon selection
- Characterization of blips from through going muons
- Conclusion

Motivation & Goal

- Muon capture at rest (muCAR) : low energy (MeV) nuclear process that produces secondary particles and associated blips that could affect low energy physics goals in DUNE

$$\mu^- + p \rightarrow n + \nu_\mu$$

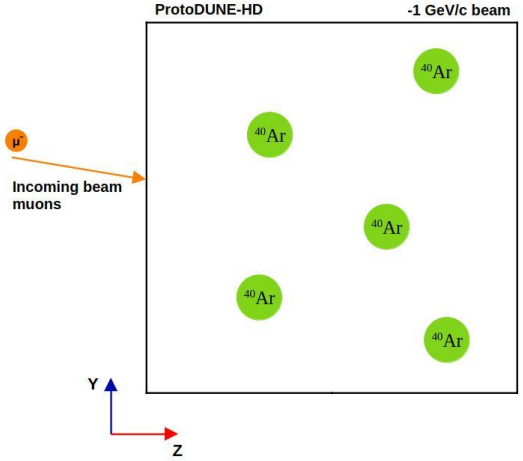
- Understanding this helps in
  - Better nuclear de-excitation models
  - μ and π discrimination using low energy signals
  - Improved background modelling
  - Better neutrino energy reconstruction in LArTPC

- Secondary particles from through going muons can act as a background for the muCAR studies

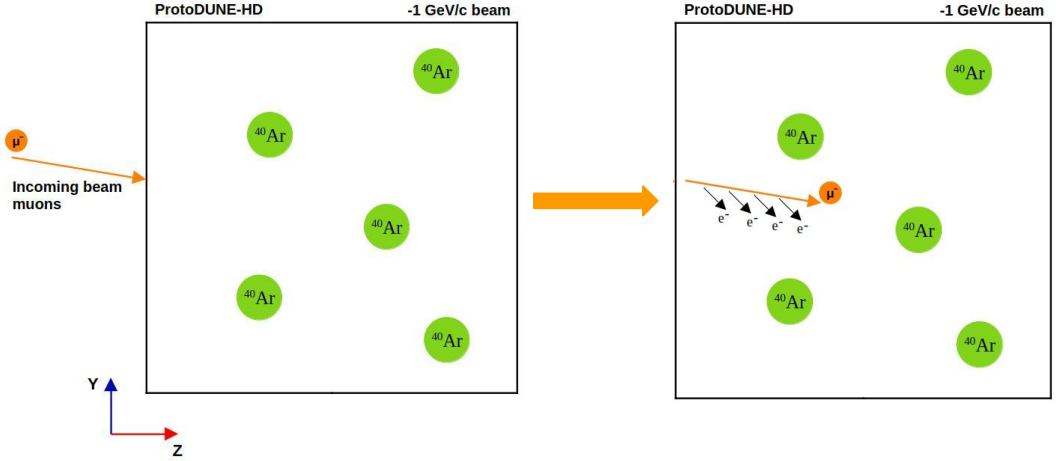
Goal

- To study the contribution of beam-induced through-going muons as a background to μCAR and to characterize the resulting blips

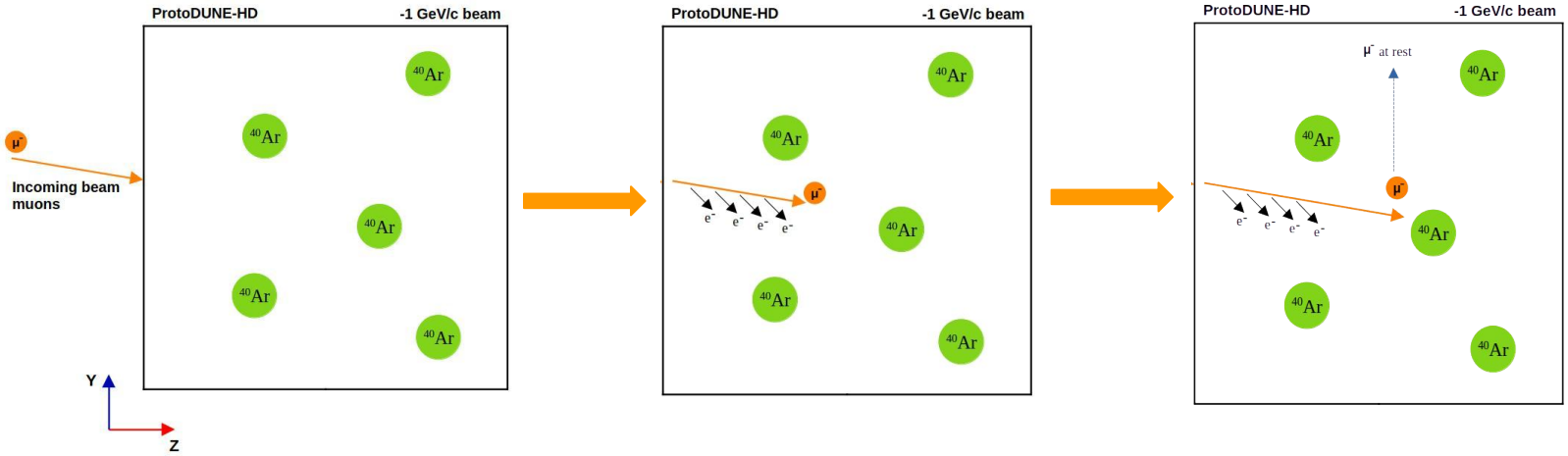
Muon Capture at Rest Process



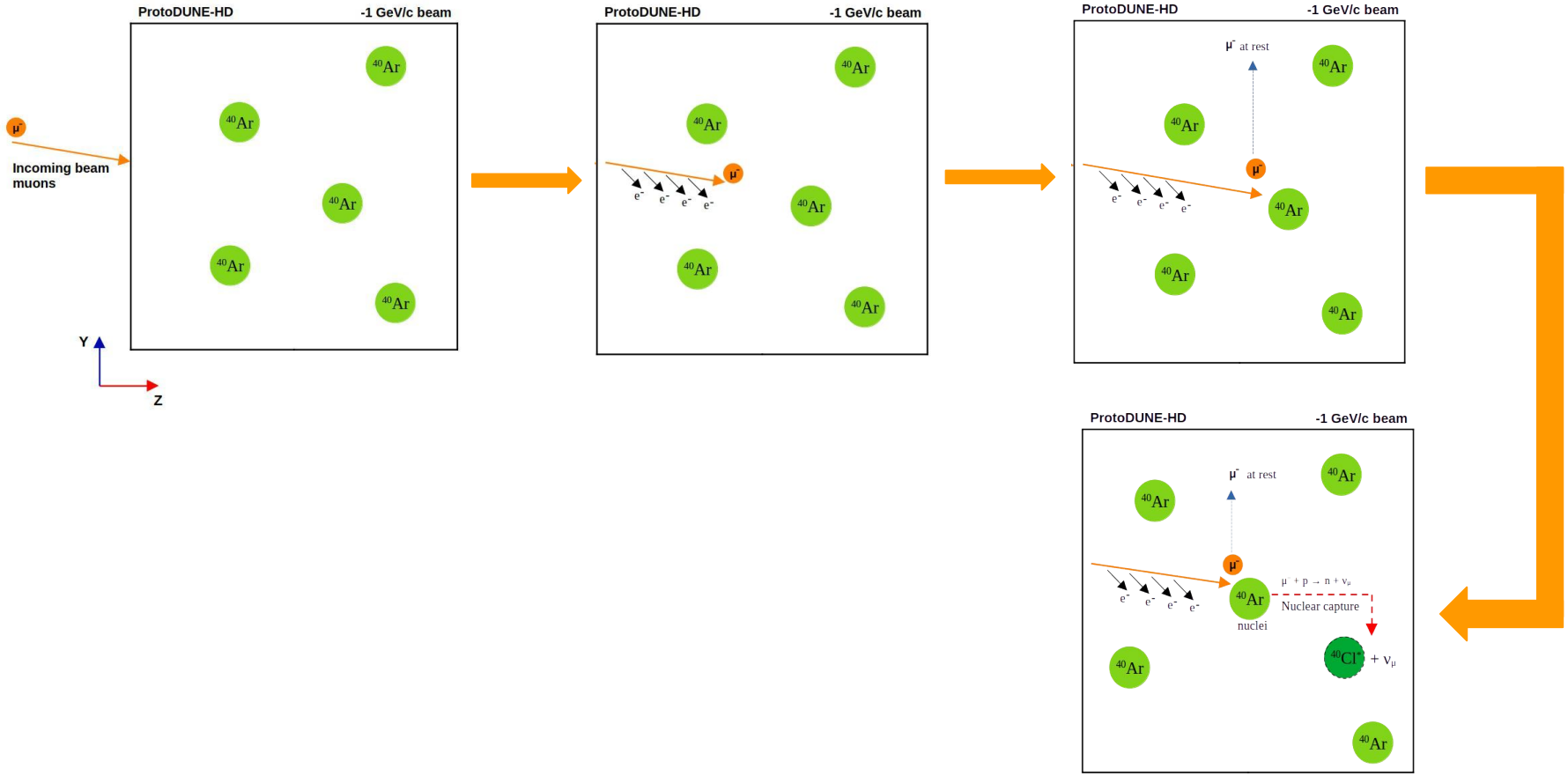
Muon Capture at Rest Process



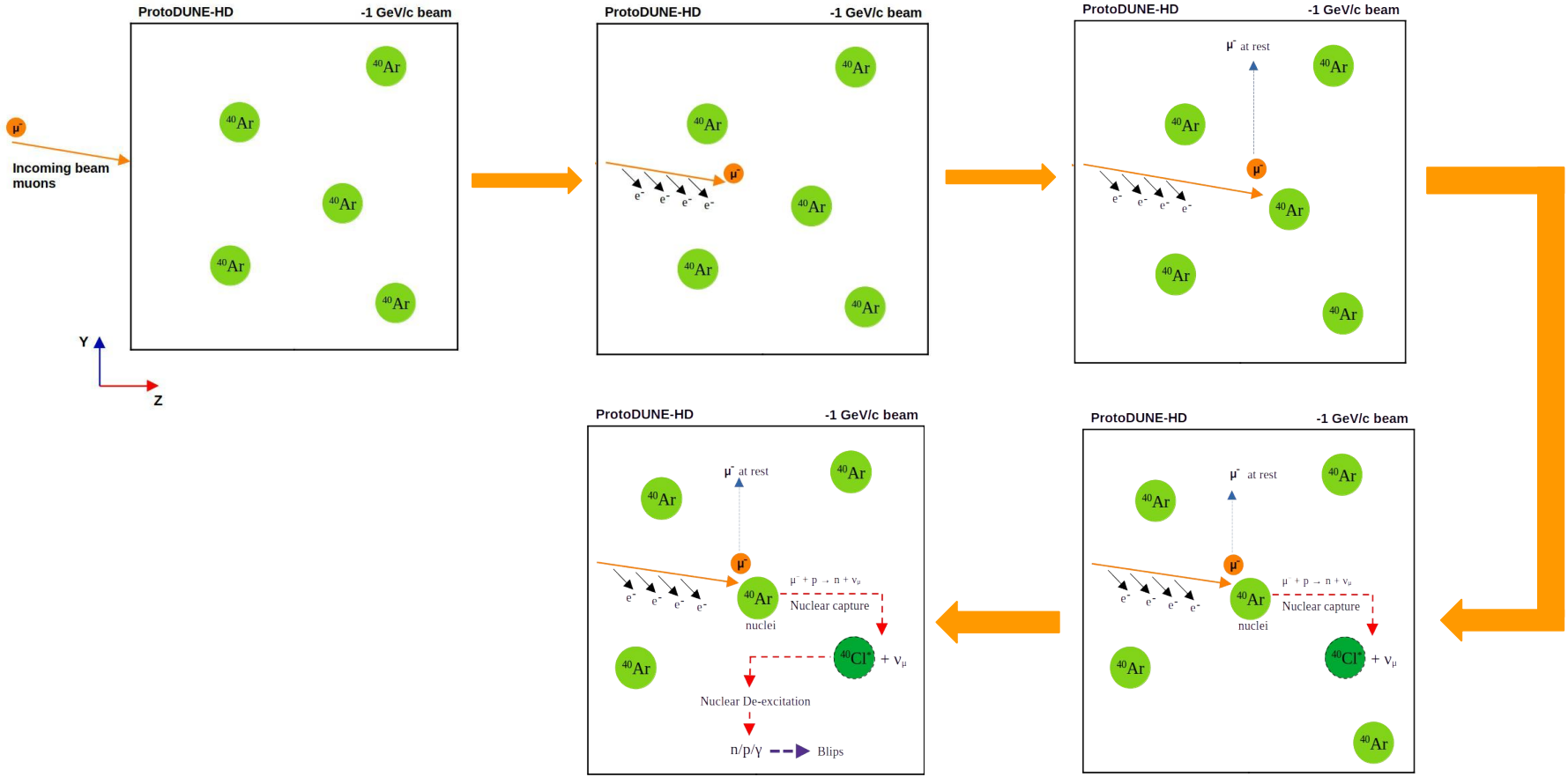
Muon Capture at Rest Process



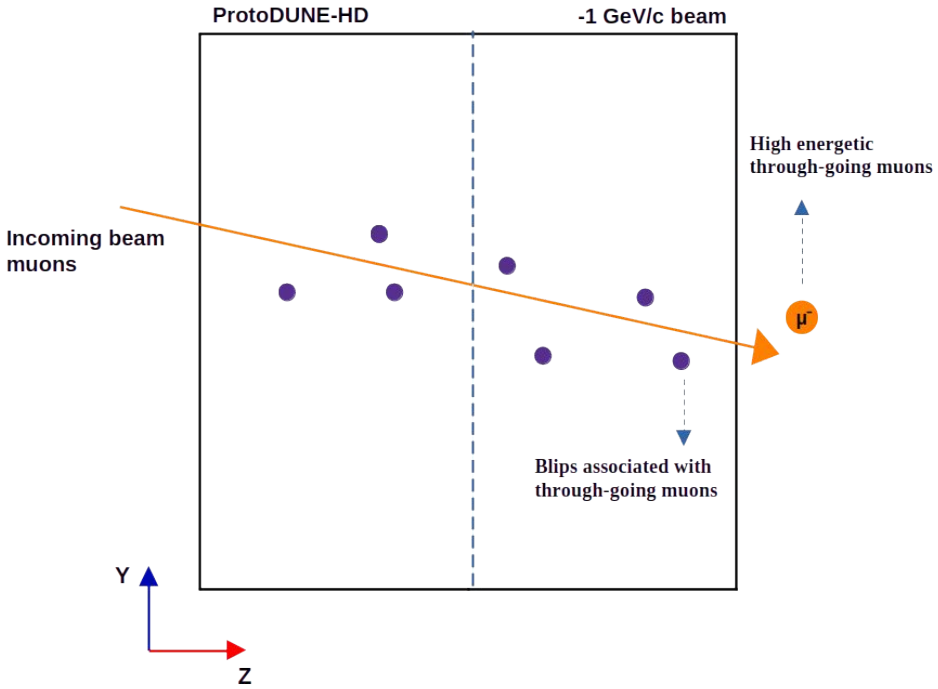
Muon Capture at Rest Process



Muon Capture at Rest Process



Through-going beam muons



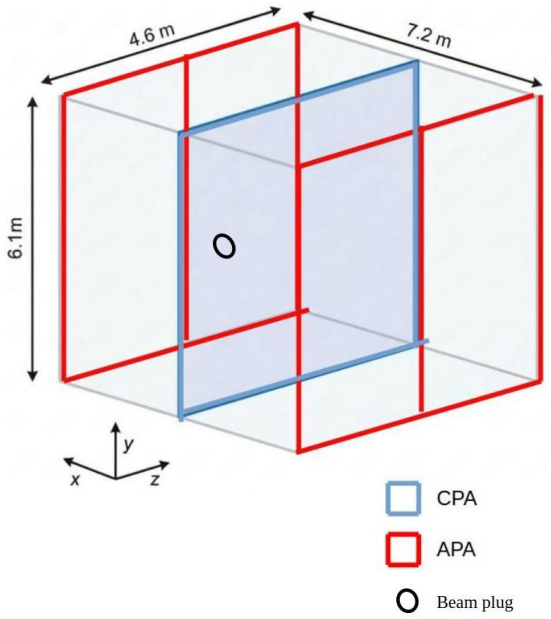
- They produce secondary particles, mainly
  - δ-electrons
  - Bremsstrahlung photons

  This generates blip like energy deposits inside the detector that act as background for muCAR

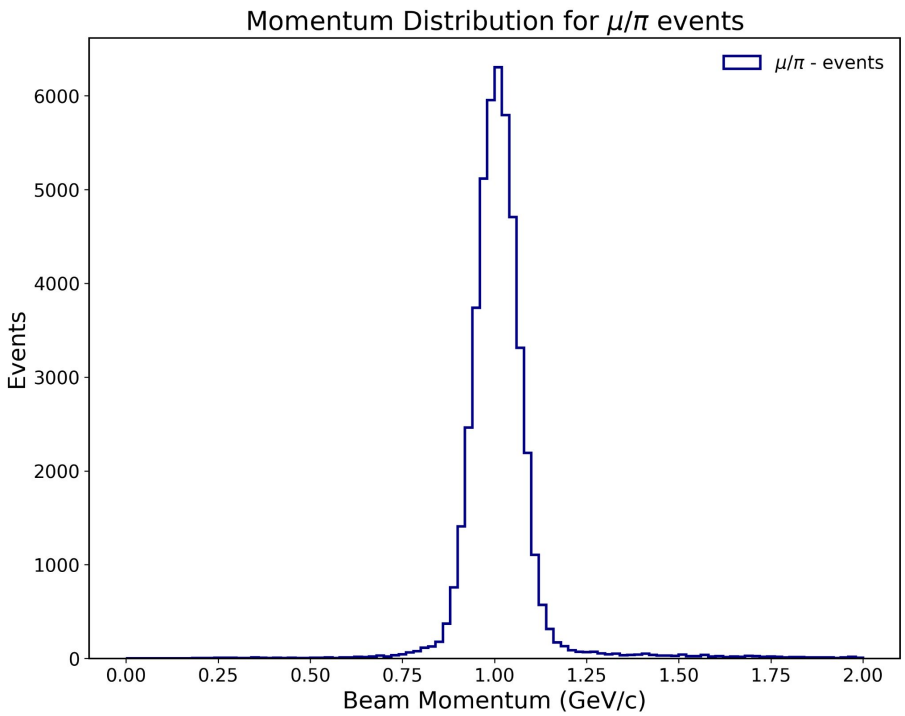
Data Used for Analysis

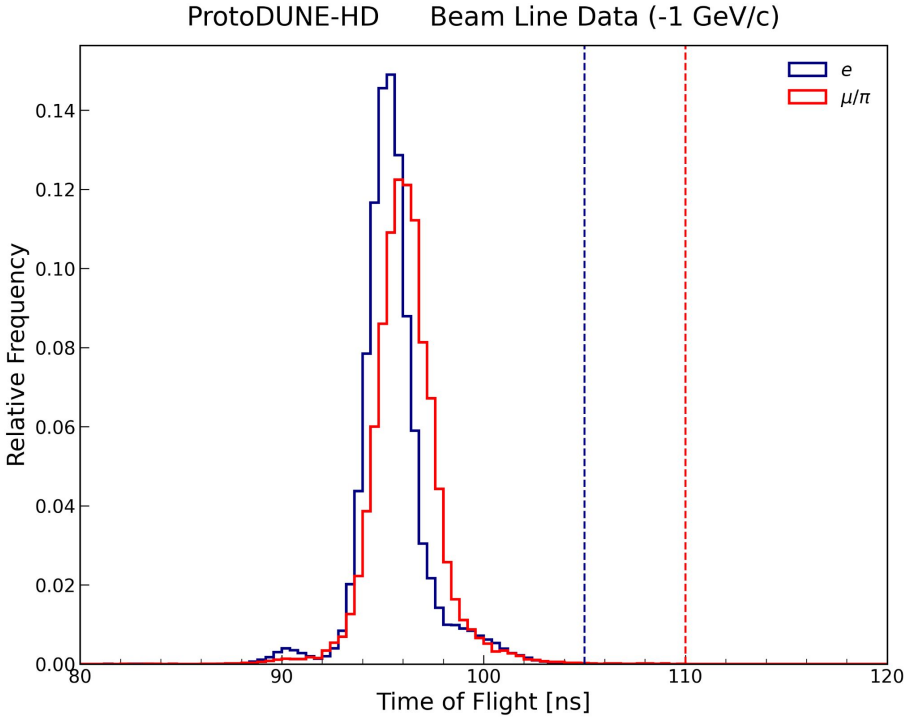
- Beam momentum : **-1 GeV/c**
- Run number : **28049**
- Total Events : **188772**
- Total μ/π events : **53210**
- μ/π events for which the beamline reconstructed and TPC reconstructed position and direction information are known : **24397**

PDHD Geometry



Momentum & TOF Distribution





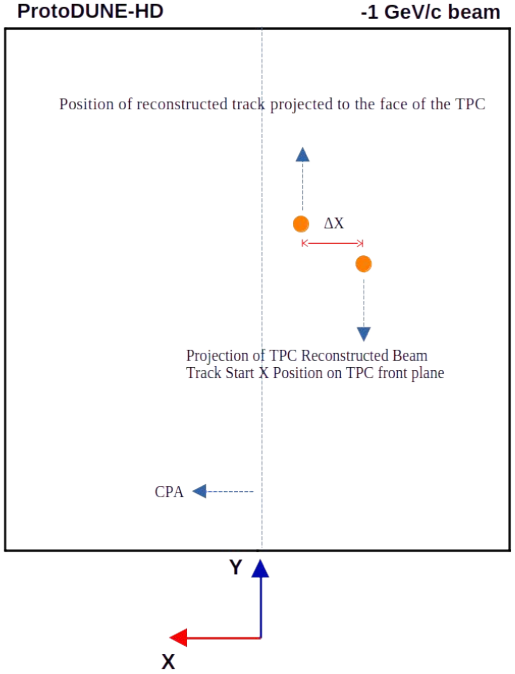
In this (-1 Gev/c) momentum the beam instrumentation is not able to seperate between pions and muons.

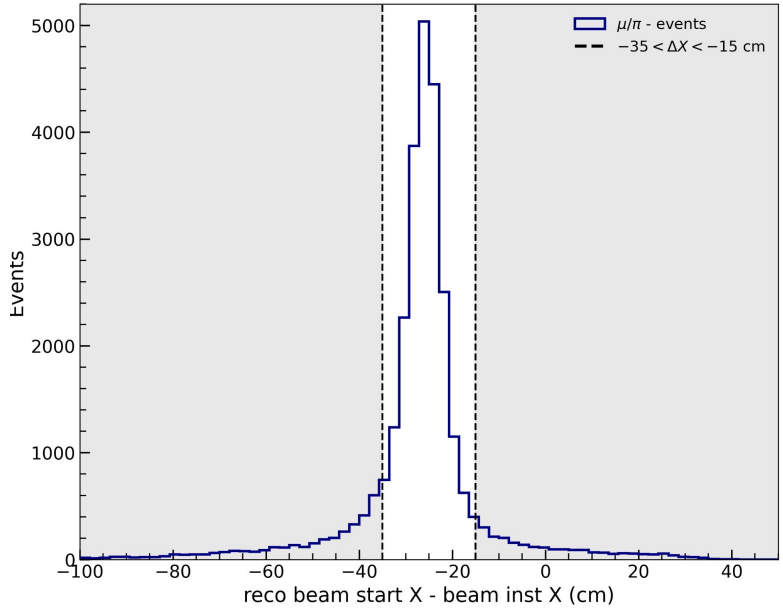
Selection of True μ/π Beam Tracks

X^{Start}_{Beam} is the start position in X of the reconstructed candidate beam track in the TPC.

X^{End}_{TPC} is the projected X location of the intersection of the track from the beam line instrumentation with the TPC

Beam quality cuts : position cuts : $-35 < X^{Start}_{TPC} - X^{End}_{Beam} < -15$



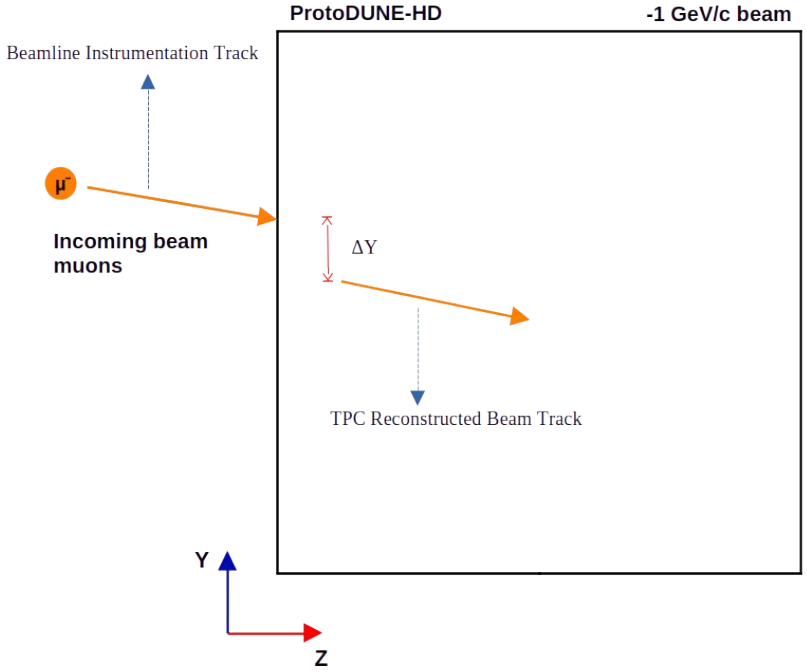


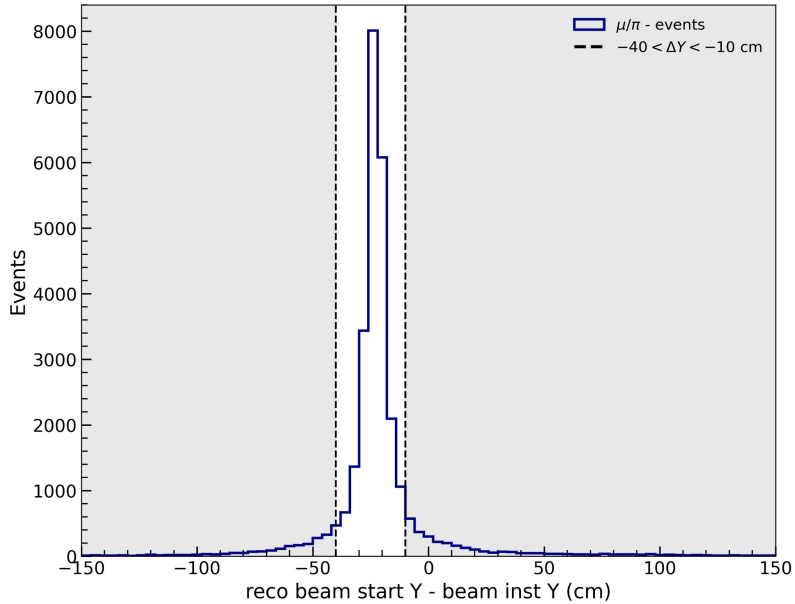
Selection of True μ/π Beam Tracks

Y^{Start}_{Beam} is the start position in Y of the reconstructed candidate beam track in the TPC.

Y^{End}_{TPC} is the projected Y location of the intersection of the track from the beam line instrumentation with the TPC

Beam quality cuts : position cuts : $-40 < Y^{Start}_{TPC} - Y^{End}_{Beam} < -10$



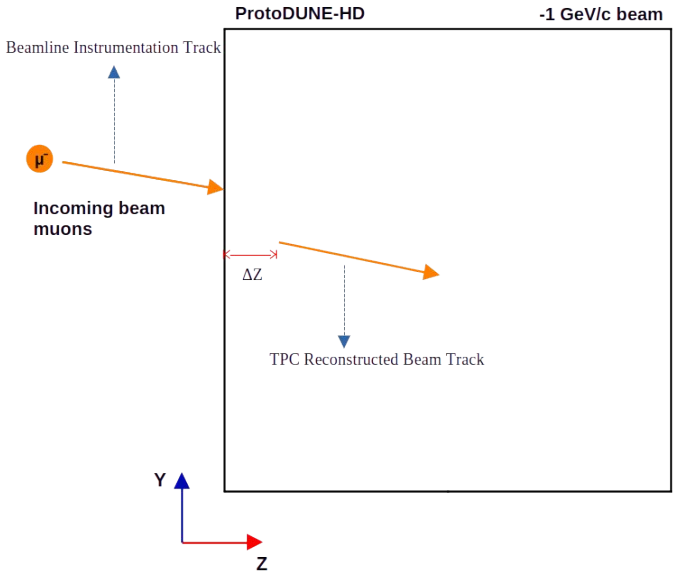


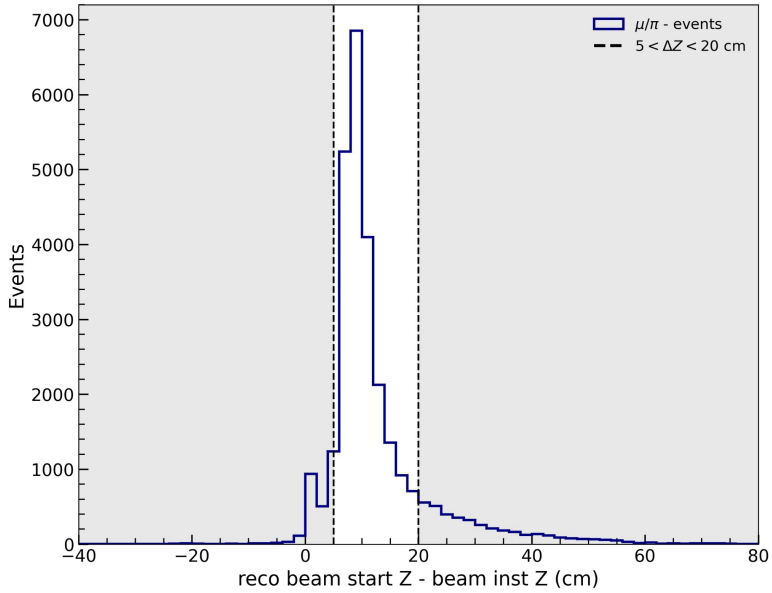
Selection of True μ/π Beam Tracks

Z^{Start}_{Beam} is the start position in Z of the reconstructed candidate beam track in the TPC.

Z^{End}_{TPC} is the projected Z location of the intersection of the track from the beam line instrumentation with the TPC

Beam quality cuts : position cuts : $5 < Z^{Start}_{TPC} - Z^{End}_{Beam} < 20$



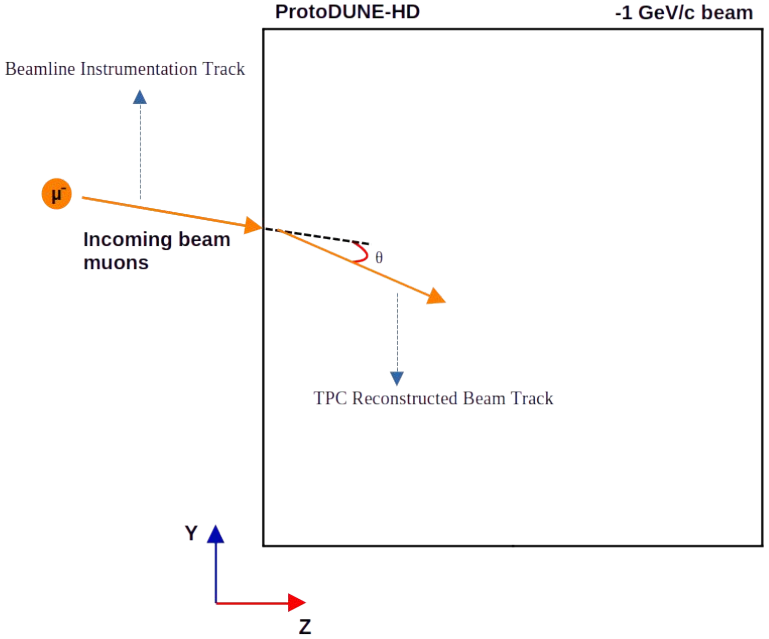


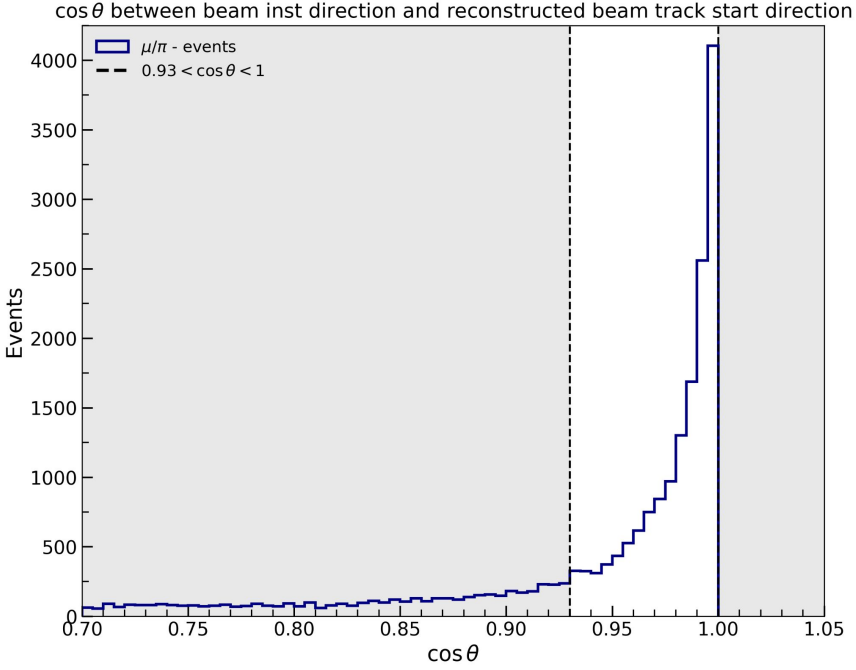
Events before X,Y,Z position cut : **24397**

Events passing X,Y,Z position cut : **15798**

Selection of True μ/π Beam Tracks

Angle cut : cos**θ** : The cosine of the angle between the beam line track and reconstructed TPC track is required to be > 0.93.



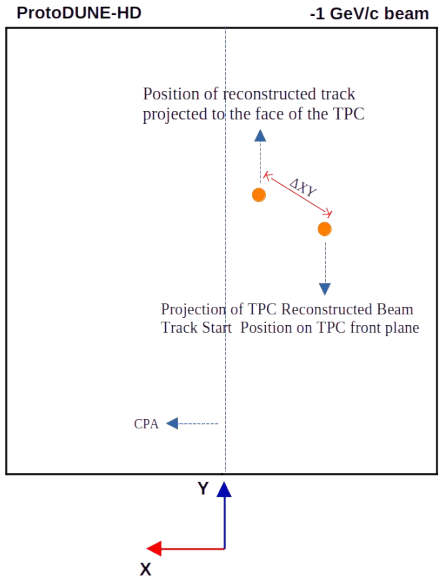


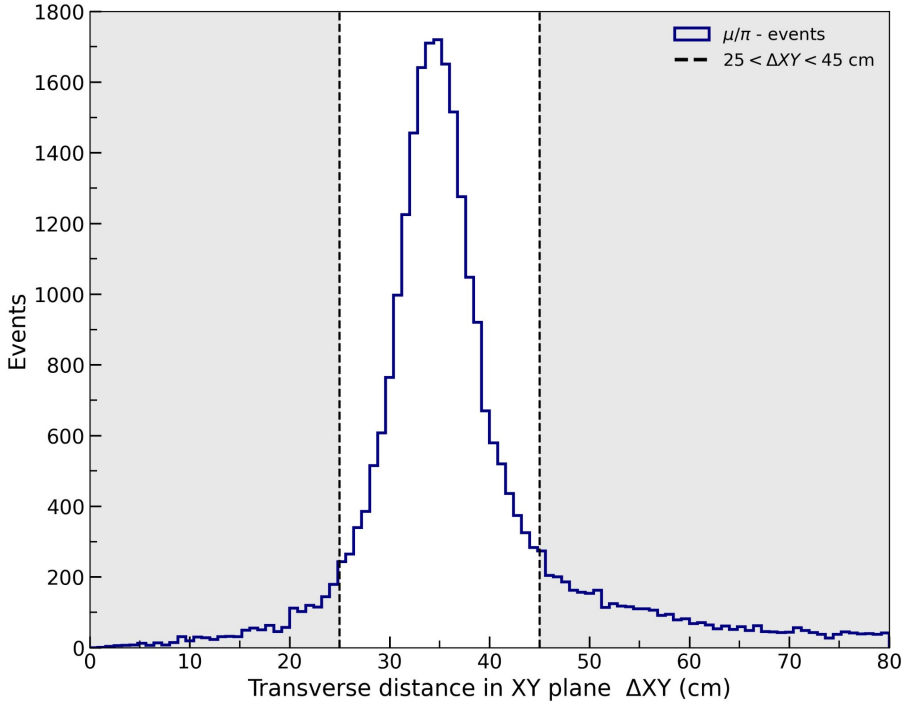
Events passing angle cut alone : **14184**

Events passing X,Y,Z position cut + angle cut : **11512**

Selection of True μ/π Beam Tracks

Transverse distance cut : Distance between the position of beamline reconstructed track projected to the face of the TPC and start position of reconstructed beam track in the XY plane



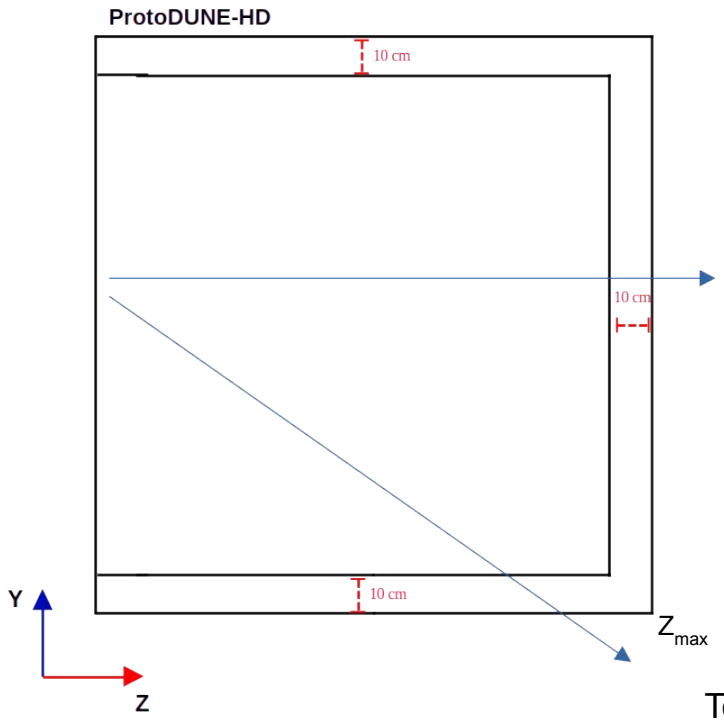


Events passing transverse distance cut alone : **18816**

Events passing X,Y,Z position cut + angle cut + transverse distance cut : **11298** ⟹ Proper beam reconstructed tracks

Selection of Through-going μ/π

Selection cut applied to obtain through-going μ/π tracks from the filtered data :



Edgecut = 10 cm

- X_{max} = 360 cm , X_{min} = -360 cm
- Y_{max} = 610 cm , Y_{min} = 0 cm
- Z_{max} = 460 cm

The limits are applied based on PDHD geometry

Selection cut :

$$[\ X^{END}_{TPC} > (X_{max} - edgecut)\ |\ X^{END}_{TPC} < (X_{min} + edgecut)\ |$$
$$Y^{END}_{TPC} > (Y_{max} - edgecut)\ |\ Y^{END}_{TPC} < (Y_{min} + edgecut)\ |$$
$$Z^{END}_{TPC} > (Z^{max} - edgecut)\ ]$$

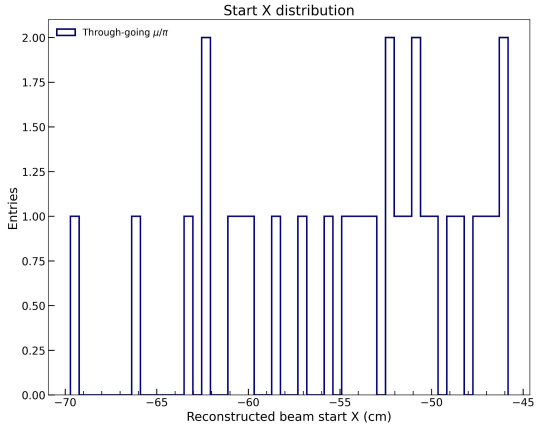
X^{END}_{TPC} , Y^{END}_{TPC} , Z^{END}_{TPC} are the position of reconstructed track projected to the face of the TPC

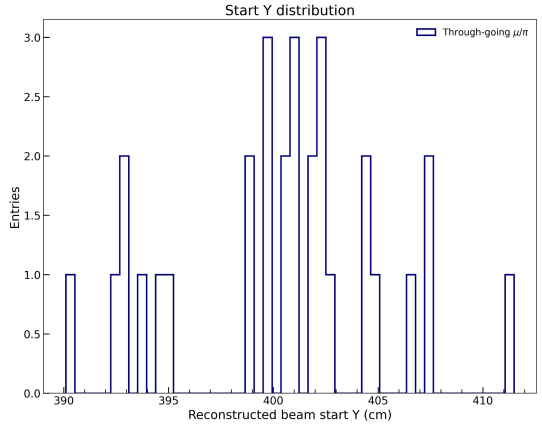
Total events passing through-going μ/π cut from the unfiltered data : **203**
Total events passing through-going μ/π cut from the filtered dataset : **30**

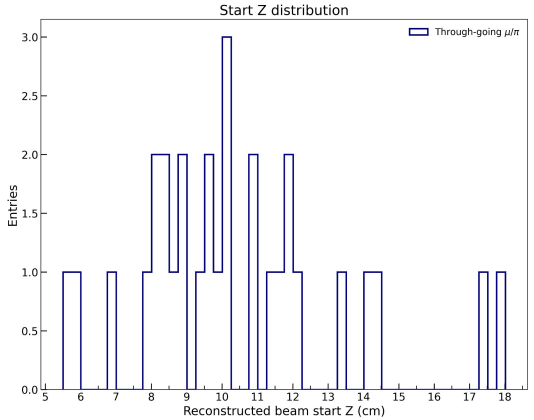
Selection Summary

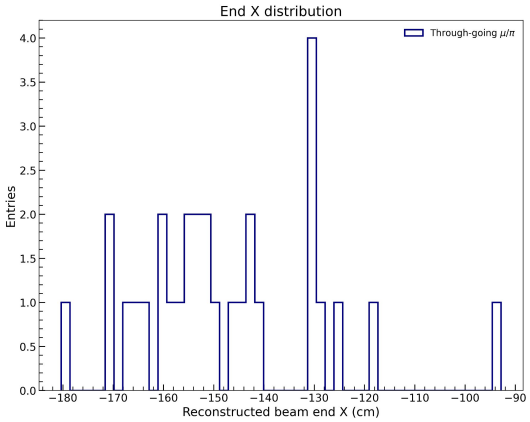
Total μ/π events	: 53210
Total μ/π events with beamline and reconstruction information	: 24397
μ/π events after X,Y,Z position cut	: 15798
μ/π events after X,Y,Z position + angle cut	: 11512
μ/π events after X,Y,Z position + angle + transverse cut	: 11298
Through going μ/π beam tracks	: 30

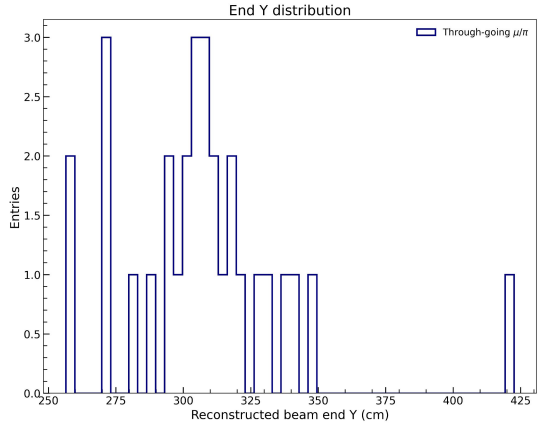
Through-going μ/π Start & End Position Distribution

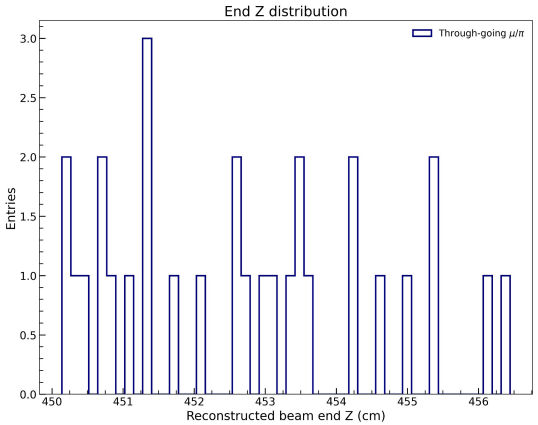




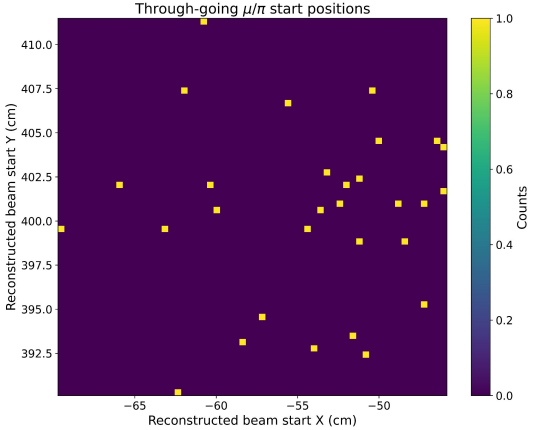


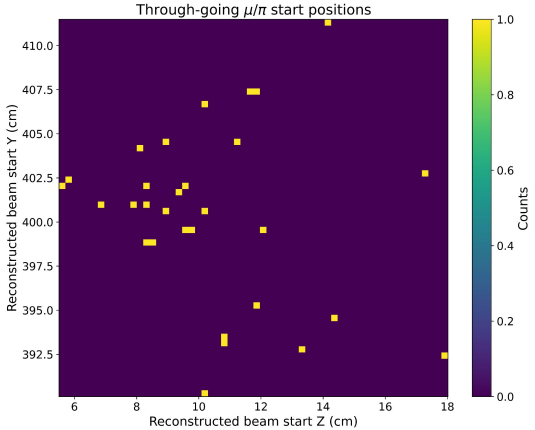


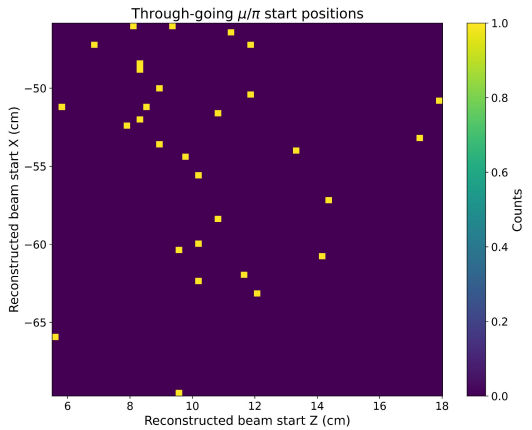


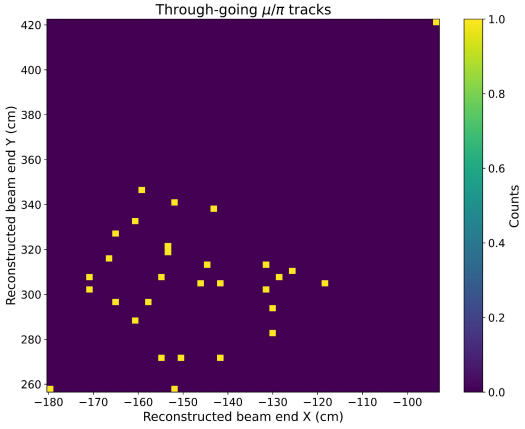


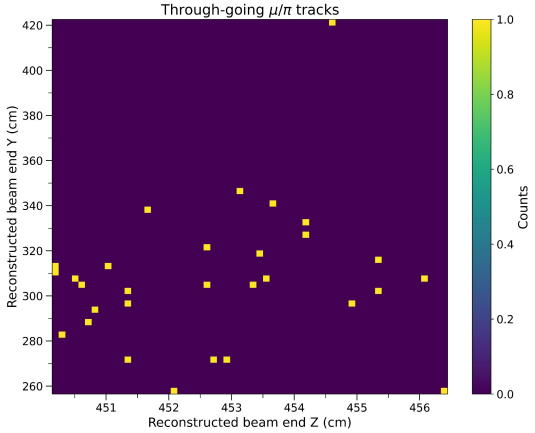
Through-going μ/π Start & End Position Distribution

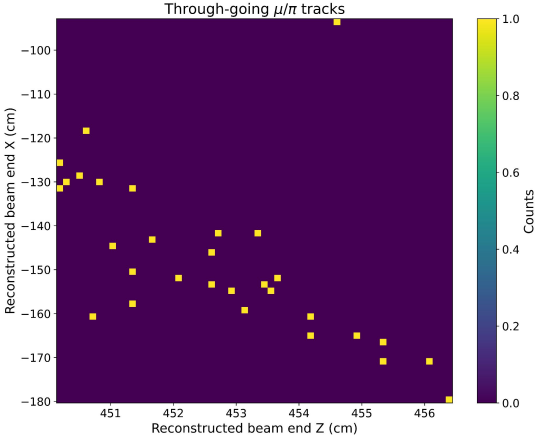




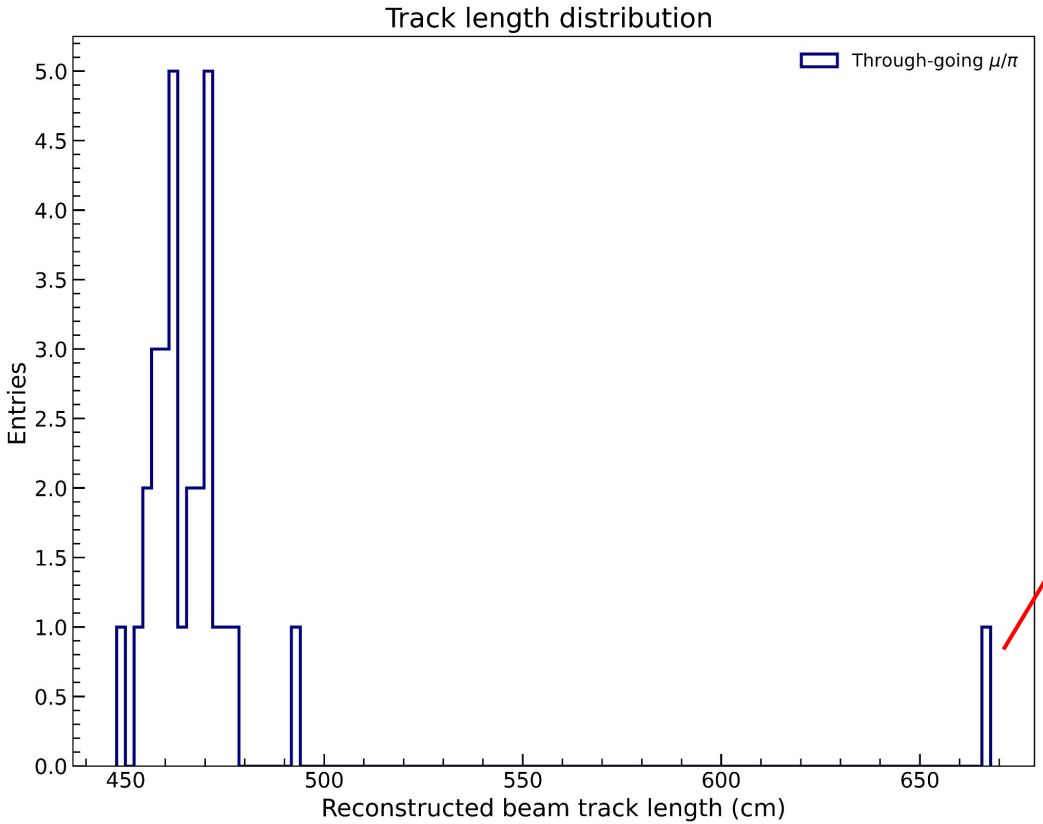








Through-going μ/π Track Length



Start (X, Y, Z) = (-60.80, 411.49, 14.16)

End (X, Y, Z) = (-92.83, 422.60, 454.61)

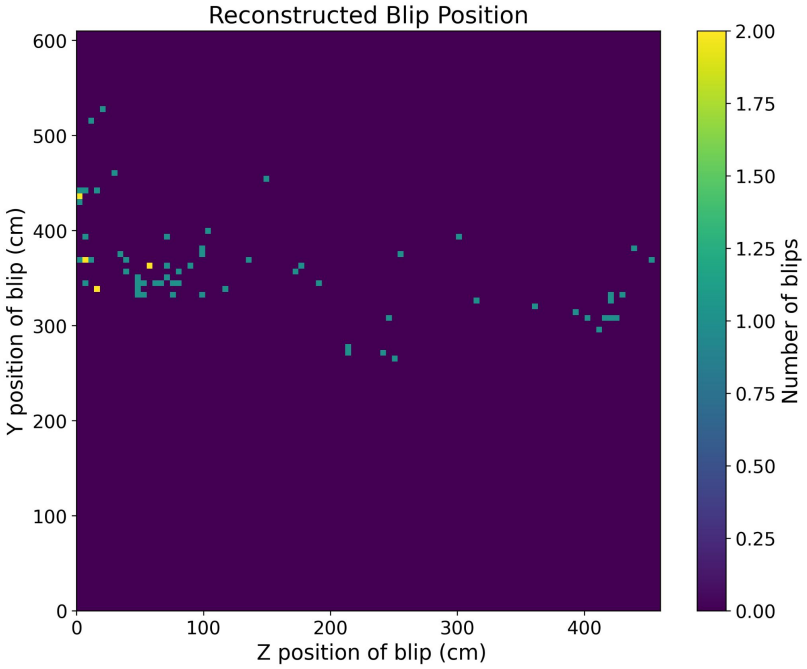
Track length = 667.83 cm

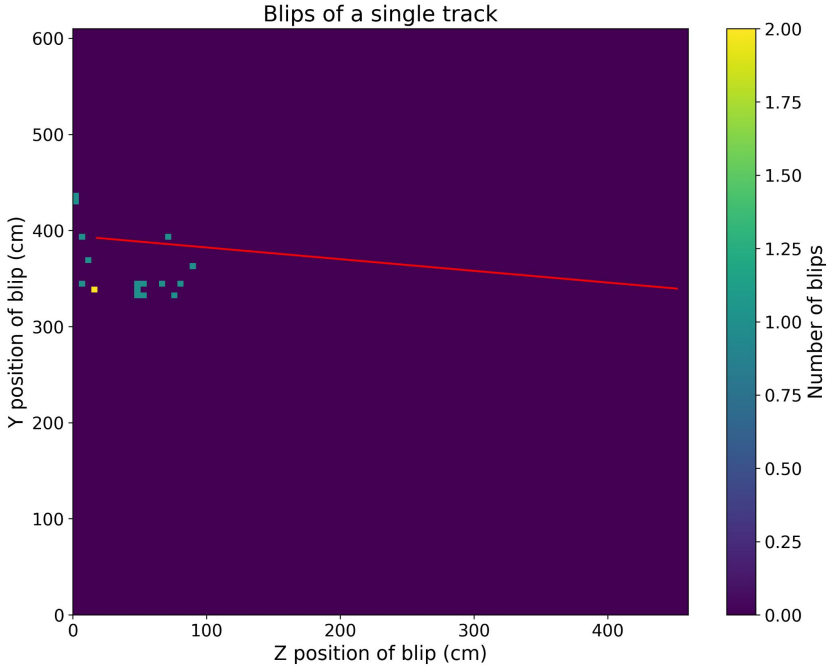
Blips

Total through-going μ/π tracks are : **30**

No of blips associated with this 30 through-going μ/π tracks are : **68**

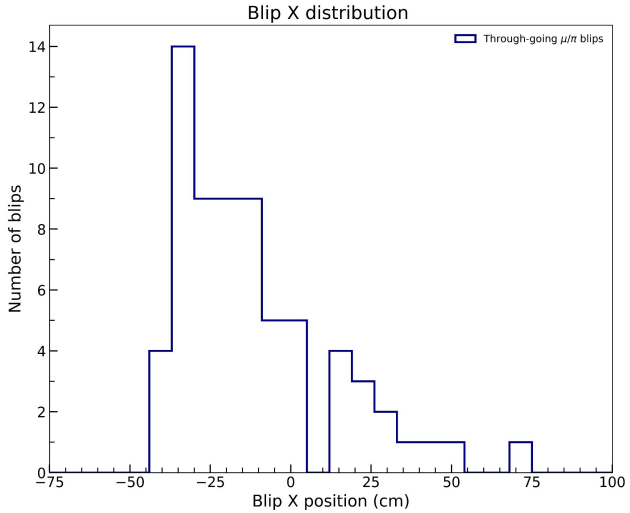
A maximum of **17** blips were found to be associated with a single track

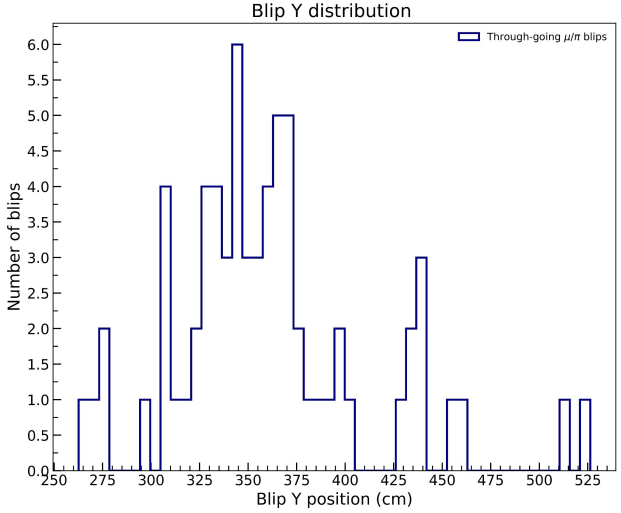


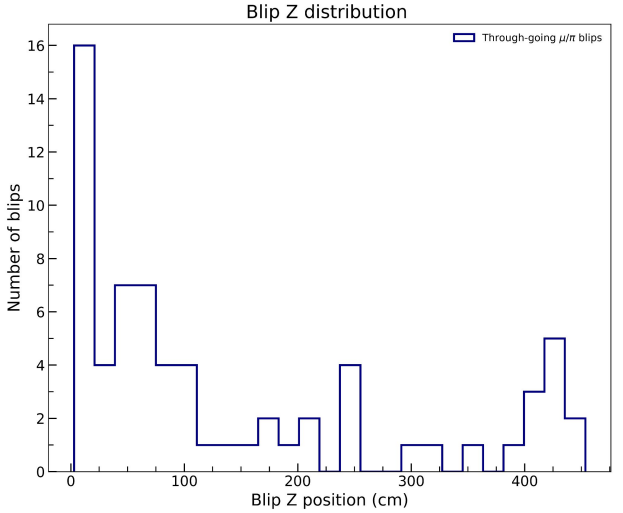


Reconstructed Blip position in PD-HD

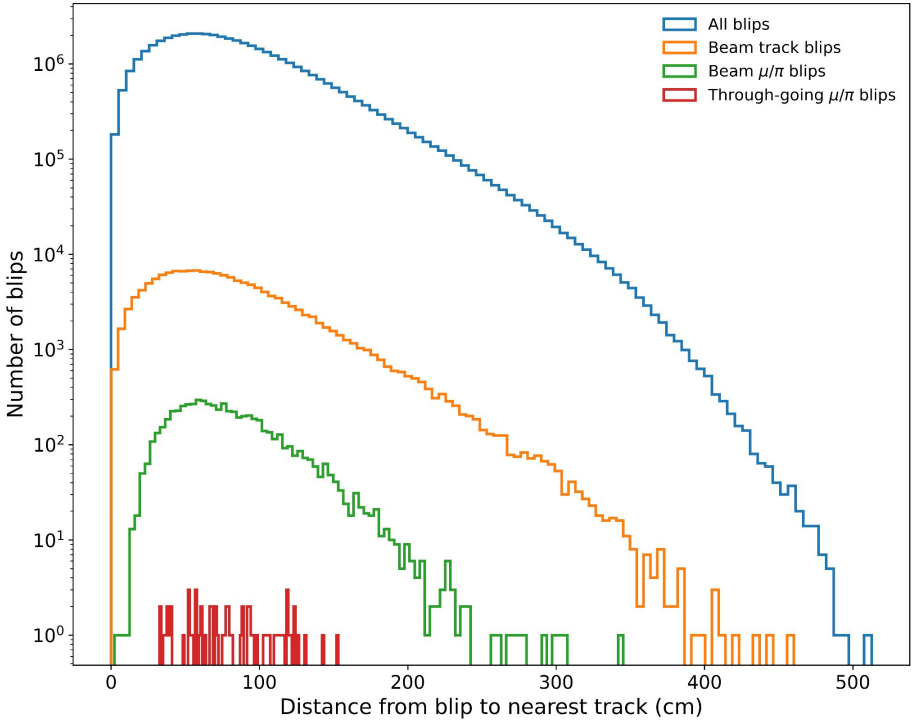
Total number of blips : **68**

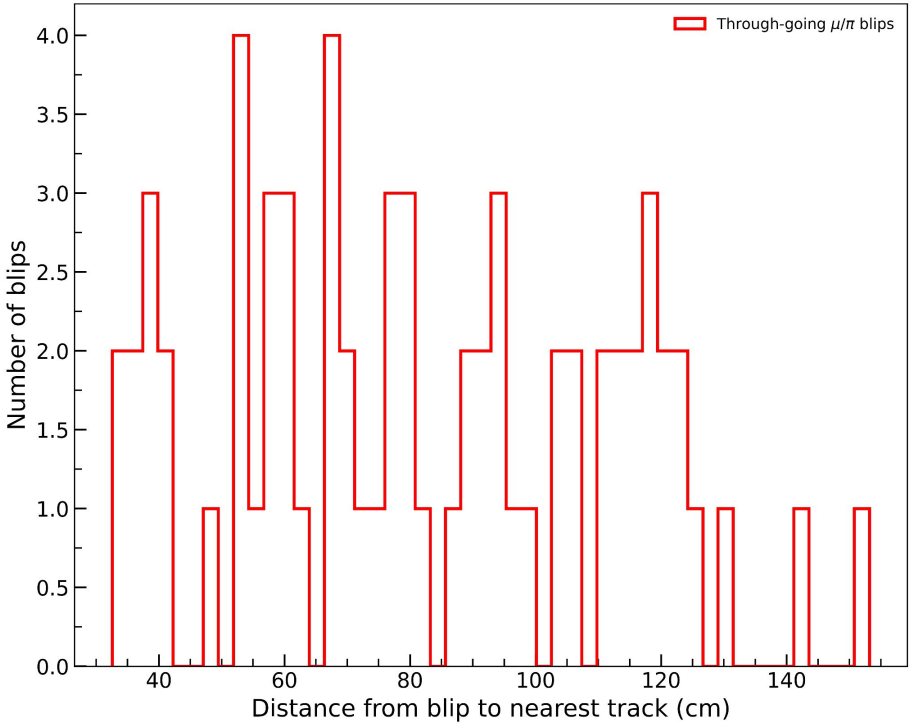




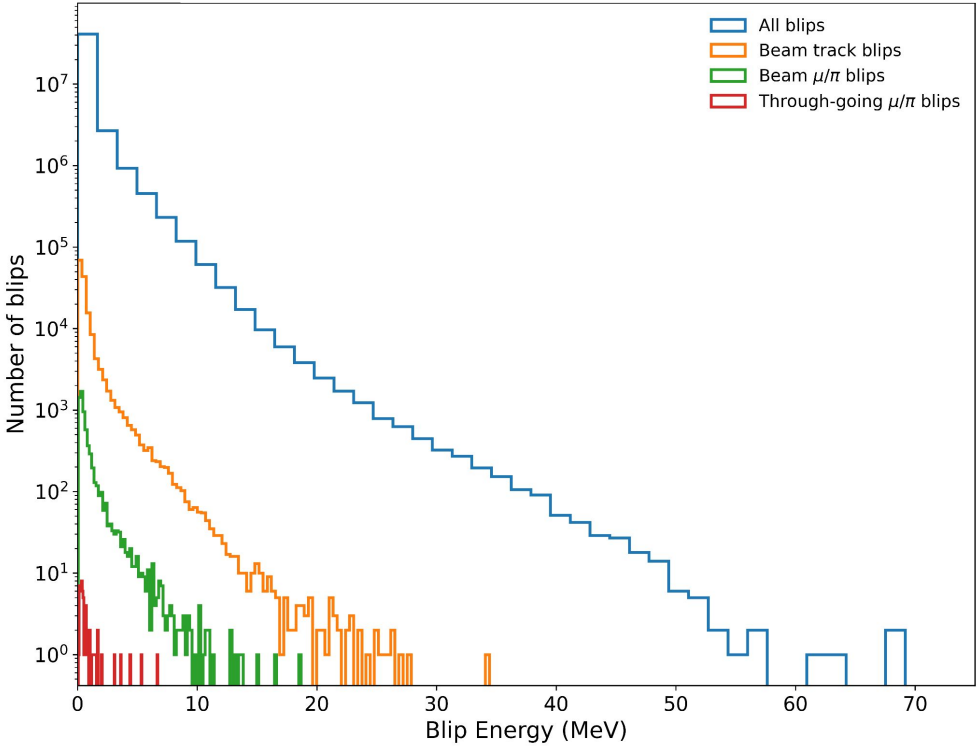


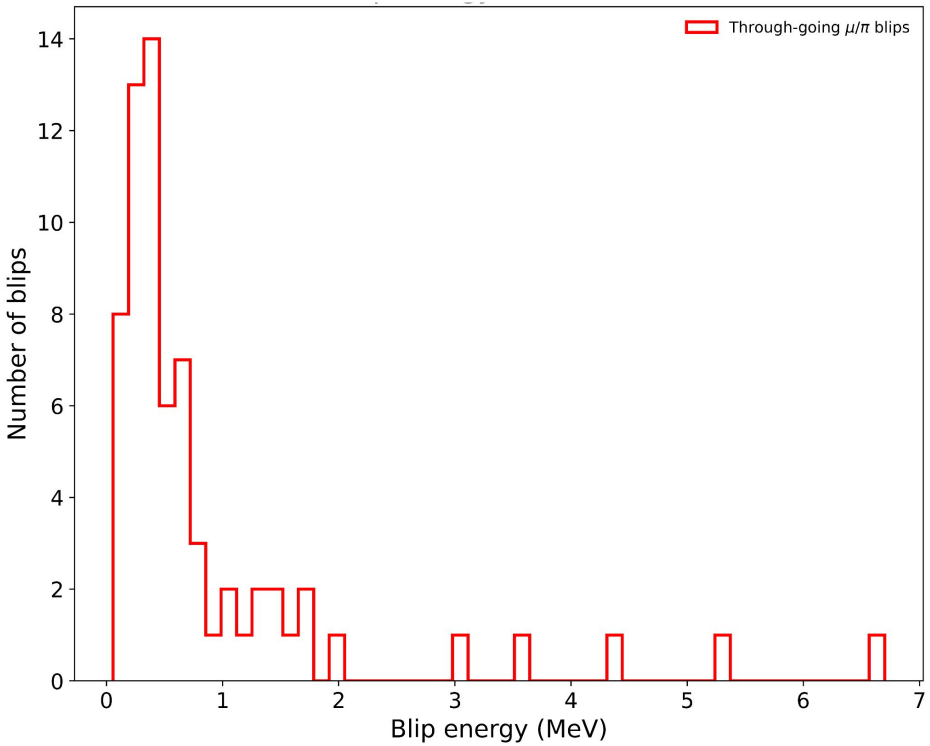
Proximal Distance of Blips to Associated Tracks





Energy of Blips to Associated Tracks





Conclusion

- **Beam Track Selection**

  Successfully identified and selected high-quality beam tracks from the reconstructed dataset using data-driven selection cuts.

- **Through-going μ/π track selection**

  Through-going beam μ/π tracks were identified, and blip reconstruction tools were implemented within the ProtoDUNE-HD framework to study signals and associated backgrounds.

- **Through-going μ/π background at -1 GeV/c beam**

  At -1 GeV/c the number of through-going beam muon/pion tracks are found to be low and the corresponding blip activity is also minimal □ the background from this sources are very low.

- **Impact on muCAR**

  muCAR studies at -1 GeV/c can be carried out without significant contamination from the through-going beam μ/π .

Thanks to Will Foreman, Bryce Littlejohn

Future work

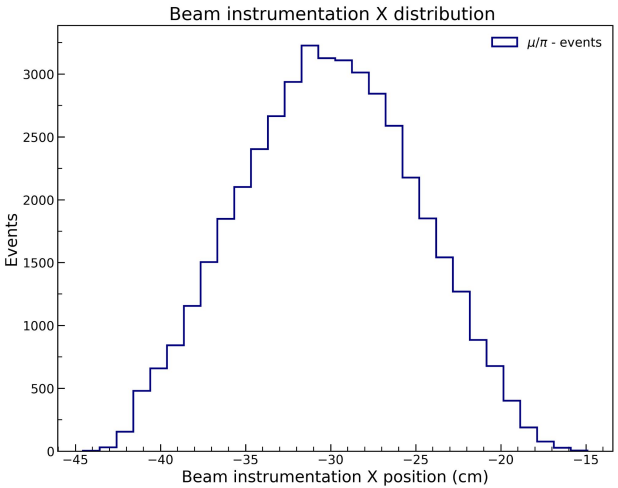
- Develop a method to separate between μ/π in an event
- Extend the study and analysis to multiple run numbers at -1 GeV/c for a better statitics
- Search for muon capture at rest and study their blip characteristics

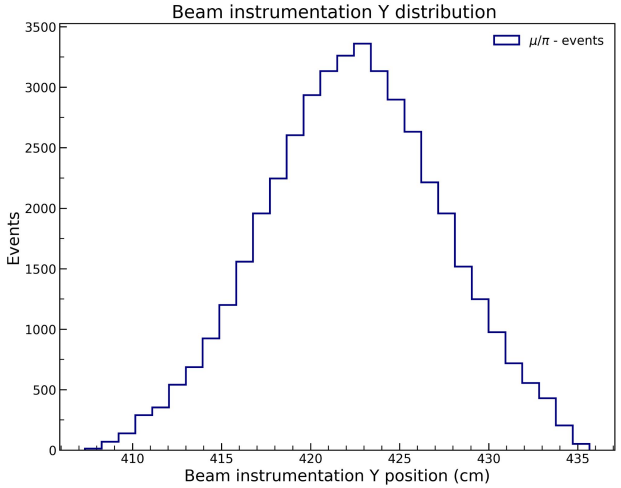
Questions ?

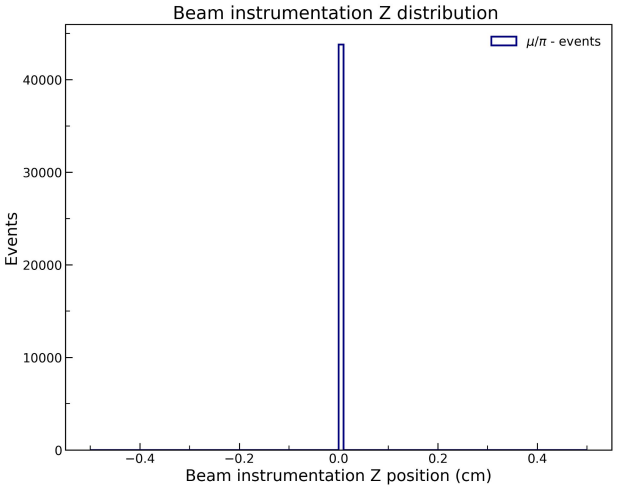
NOTE : I am looking for PhD opportunities

adikirann@gmail.com

Backup


Beam instrumentation X distribution
μ/π - events
Events
Beam instrumentation X position (cm)


Beam instrumentation Y distribution
μ/π - events
Events
Beam instrumentation Y position (cm)


Beam instrumentation Z distribution
μ/π - events
Events
Beam instrumentation Z position (cm)